

AMANDEEP

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **6+ years** architecting distributed, event-driven systems in Python and Java for **Insurance and Fintech**; owning system design, SOC 2-compliant data architecture, and full delivery cycles end-to-end. Drove **\$3M+ in new ARR**, cut infrastructure costs **\$200K annually**, shipped AI-powered platforms saving **1,000+ engineering hours**, and reduced **SLA violations 70%**.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, Go, C, C++, Bash, Shell

Backend & Frameworks: FastAPI, Flask, Django, Spring Boot, Hibernate, gRPC, React

Cloud & DevOps: AWS (EC2, ECS, RDS, S3, SQS, Lambda), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins)

Data & Messaging: PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, BigQuery, Kafka, Google Pub/Sub

Monitoring: Datadog, Prometheus, Grafana, Elasticsearch

Practices: Microservices, Event-Driven Architecture, ETL, Distributed Systems, TDD, Observability

AI-Assisted Development: Development, testing, debugging, and technical design using Claude and Codex

WORK EXPERIENCE

Senior Software Engineer | **Independent** | Remote, US

Oct 2025 – Present

- Architected **MatchKinetics AI**: a decoupled job intelligence platform with automated discovery, structured extraction, and a crawler combining direct ATS APIs (**GraphQL/REST**) and **Playwright** for JS-rendered SPAs, processing **300+** pages daily across **8 platforms**.
- Implemented high-throughput parallel LLM pipelines using OpenAI (**GPT-4o**) and local **Ollama** with thread locks for SQLite, achieving sub-second structured data extraction with **95%+** classification accuracy across **10,000+** records.
- Engineered a fault-tolerant crawling infrastructure with exponential backoff for **rate-limited APIs**, bot-detection-aware request strategies for **Cloudflare**-protected endpoints, and **JSON-LD** structural parsing fallbacks: sustaining **95%** crawler reliability end-to-end.
- Built a **React/TypeScript** interface over a **FastAPI** backend, enabling real-time filtering and ranking of **10,000+** job listings with sub-second query response.

Senior Software Engineer | **Afficiency** | NYC, US

Jan 2024 – Sept 2025

- Delivered a company-wide **internal automation platform** (Python/Flask + FastAPI, React) adopted across engineering, QA, and product, with a request-replay system running **3,000+ test cases** in one click and a real-time audit dashboard for API payload inspection, cutting manual testing **70%** and saving **1,000+ engineering hours**.
- Developed an **AI-powered log-summarization system** (Python, Ollama, Mistral) integrated with Microsoft Teams and Elasticsearch to proactively detect API failures and auto-generate contextual summaries, reducing **SLA violations by 70%**.
- Designed an event-driven **automated underwriting system** on **Kafka** processing **10,000+ policies annually (\$2.5M+ ARR)** and reduced decision time **80%**; integrated 6+ third-party compliance and data services (HIPAA, MIB, LABIQ, EHR) behind REST APIs with Redis caching for sub-second responses.
- Designed a **multi-tenant data isolation architecture** (AWS RDS/MySQL + S3) to meet **SOC 2** and carrier-grade security requirements, enabling shared infrastructure across clients while enforcing strict per-tenant data boundaries.
- Led a zero-downtime **AWS-to-Hetzner migration** that cut infrastructure costs **~\$200K annually**, and established a reusable migration framework that reduced subsequent migrations from 4 weeks to 1 week; hardened **Docker/Kubernetes** orchestration with a **Prometheus/Grafana** observability stack and **HashiCorp Vault** for secrets management.
- Led a team of **5+ engineers** to deliver a production-ready carrier integration in 2 weeks, securing a **\$3M+ ARR** contract.

Software Engineer | **Capgemini** | NYC, US

Jul 2021 – Dec 2023

- Engineered an enterprise **Underwriting Management System** (Spring Boot microservices + **Python ETL**) handling **500+** concurrent transactions at **99.9% uptime**, built on **12+** microservices with Hibernate, PostgreSQL, and Redis.
- Integrated external insurance data providers via **REST/gRPC** with schema validation and contract enforcement; optimized Jenkins/GitHub Actions CI/CD for **40% faster deployments** and introduced **Datadog APM** tracing that cut **MTTR ~40%**.
- Designed and enforced a **multi-layer test strategy** (unit, integration, and functional) with **pytest** across **12+** microservices, backed by automated code-quality gates (pylint, black) in CI/CD; reducing production defects **30%**.

Software Engineer | **ATEC Wireless** | Remote, US

Jan 2020 – May 2020

- Built a **Python-based ML recommendation engine** (scikit-learn K-Means clustering) to optimize mobile-tower signal coverage by processing network telemetry through automated ETL pipelines to achieve a **40%** improvement in network KPIs.
- Engineered a batch-processing pipeline with automated data-quality enforcement and **pytest** validation for **100+** tower sites.

Software Engineer | **Perennial Systems** | Pune, India

Aug 2017 – Jul 2018

- Built **Python/Java** backend services for an analytical dashboard (**Spring Boot REST + MySQL**) serving **1,000+ concurrent users**.

EDUCATION

M.S. in Computer Science

New Jersey Institute of Technology, Newark, NJ · Dec 2020

Teaching Assistant: Data Structures & Algorithms, Introduction to Artificial Intelligence

B.E. in Computer Science

Dayananda Sagar College of Engineering, India · Jul 2017